Faculty of Industrial Engineering

INGB 417
Facilities Design
Semester Group Assignment

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2. Background

Wiring.com specializes in manufacturing various steel wire netting products, including Bird, Chicken, Pig, Jackal, and Welded Mesh netting. Rising demand has outgrown the company's current facility, and expansion on-site is not feasible due to space constraints. To address this, Wiring.com has acquired a 50-hectare site within 10 km of its current location in Sedibeng District, Gauteng.

This project focuses on designing a new manufacturing facility to accommodate increased production and future growth. The facility will process steel coils into netting products through wire drawing, galvanization, netting formation, and packaging. With a projected demand of 26,000 tons per year, the design must optimize material flow, enhance efficiency, and incorporate sustainable energy solutions.

The project involves conceptual layout planning, considering operational needs, logistics, and scalability. Management expects a comprehensive design, including equipment selection, process flow mapping, storage solutions, and industry best practices.

3. Design Objectives

The following design objectives, categorized under their relevant headings, have been established to ensure the successful design of Wiring.com's new manufacturing facility:

- **Production**:
  To design a manufacturing facility capable of meeting the projected annual demand of netting products, with the flexibility to accommodate for future growth.

- **Site, Layout and Flow**:
  The facility layout must minimize material handling, reduce bottlenecks, and ensure a logical flow on a greenfield site.

- **Equipment and Machinery**:
  To select state-of-the-art machinery and equipment that meet production requirements, ensure product quality, and promote worker safety, while minimizing downtime and maintenance needs.

- **Environmental and Regulatory Compliance**:
  To ensure the facility complies with environmental regulations and safety standards, while minimizing environmental impact and maintaining a safe working environment.

- **Sustainability**:
  To incorporate sustainable energy strategies, such as renewable energy sources and energy-efficient technologies.

4. Design Requirement Statements

The following design requirements, categorized under their relevant headings, have been established to ensure the successful design of Wiring.com's new manufacturing facility:

- **Production**:
  - The facility must produce a minimum of 26,000 tons of netting products annually.
  - The facility is required to produce 4,000 tons of bird netting, 5,500 tons of chicken netting, 2,500 tons of jackal netting, 8,000 tons of pig netting, and 6,000 tons of welded mesh per annum.
  - Must accommodate multiple wire diameters (0.6mm, 0.9mm, 1.8mm, 2.5mm) and netting variations.
  - Must always have enough raw material on hand to meet three days of demand.
  - 4% scrap rate must be factored into for the first operation.
- **Site, Layout and Flow**:
  - The total site size for the greenfield facility is 50 hectares, with 75% (37.5 hectares) allocated for buildings, production space, and storage, while 25% (12.5 hectares) is designated for roads, deliveries, parking, and green buffer zones.
  - Raw materials must be stored close to operations for better workflow, with a designated storage area capable of holding at least three days' worth of production demand. The storage area must be easily accessible for the weekly supplier deliveries.
  - There needs to be an area designated for the scrap produced in the first few processes.
- **Equipment and Machinery**:
  - The facility must include wire drawing machines, galvanizing baths (with no mass change due to galvanization), netting machines, and automated packaging systems.
  - All machines must be capable of handling production variations while being safe for employees' usage.
- **Environmental and Regulatory Compliance**:
  - The facility must comply with zoning, emissions, and industrial safety regulations.

- Proper waste disposal systems must be in place for scrap material and chemical byproducts from galvanization.

- **Sustainability**:
    - At least two sustainable energy strategies must be integrated into the facility, such as solar power and energy-efficient equipment.

5. Assumptions and Constraints

Table 1 outlines some constraints and assumptions made throughout the design process. Other relevant assumptions are mentioned in the document where applicable.

Assumption/Constraint	**Details**
Working Days	The facility operates 220 working days per year, with each day defined as a full shift from 8 am to 5 pm, including a 1-hour lunch break from 12 pm to 1 pm.
Galvanization Mass Effect	Galvanization has no marginal effect on mass.
Unit of Product	1 unit is equivalent to 1 roll of product.

Table 1: Assumptions and Constraints

6. Detail Design Specifications

6.1 Product Specifications

Wiring.com offers five types of wire fencing: Bird, Chicken, Jackal, Pig, and Welded Mesh netting. The details of each product are provided in the *table 2*.

PRODUCT SPESIFICATIONS						
Product	**Height of roll (m)**	**Aperture (mm)**	**Wire Diameter (mm)**	**Roll Length (m)**	**Weight per roll (kg)**	**Galvanized finished (Light or Heavy)**
Bird Netting	0,9	13	0,6	50	22,5	Light
Chicken Netting	1,8	25	0,9	50	41,9	Light
Jackal Netting	0,6	75	1,8	50	19,5	Light
Pig Netting	0,9	90	1,8	30	14,7	Light
Welded Mesh	2	50	2,5	30	70	Heavy

Table 2: Product Specifications

Figure 1 provides a visual representation of the products, with measurements shown to scale.

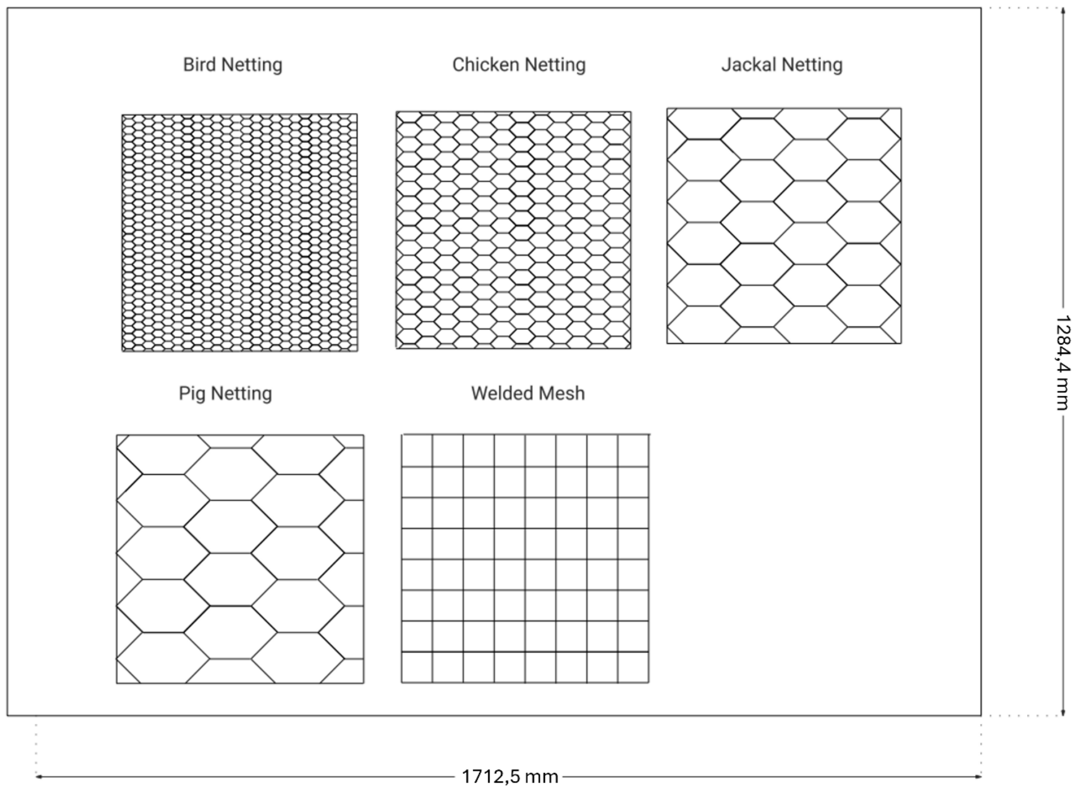

Figure 1: Detailed Visual Product

6.2 Daily Volumes

6.2.1) Raw Material Volumes

Table 3 outlines the daily raw material of steel coils required, totalling 150 coils—50 coils to meet demand and 100 coils allocated for safety stock.

RAW MATERIAL VOLUMES					
Product	**Annual Demand (tons)**	**Daily Demand (tons)**	**Daily Scrap (tons)**	**Total Daily Demand (tons)**	**Number of coils needed for production per day (coils)**
Bird Netting	4000	18,18	0,73	18,91	7,56
Chicken Netting	5500	25,00	1,00	26,00	10,40
Jackal Netting	2500	11,36	0,45	11,82	4,73
Pig Netting	8000	36,36	1,45	37,82	15,13
Welded Mesh	6000	27,27	1,09	28,36	11,35
TOTAL	26000	118,18	4,73	122,91	50,00
Safety Stock	100,00				
Total Steel Coils on Hand Per Day (rounded up)	150				

Table 3: Raw Materials Calculations

6.2.2) Finished Product Volumes

Table 4 presents the total number of units required per product: 809 rolls of Bird netting, 597 rolls of Chicken netting, 583 rolls of Jackal netting, 2474 rolls of Pig netting, and 390 rolls of Welded Mesh netting. This results in a total daily requirement of 4853 units.

FINISHED PRODUCTS				
Product	**Annual Demand (tons)**	**Daily Demand (tons)**	**Daily Demand (kg)**	**Daily Number of Units** (rounded up)
Bird Netting	4000	18,18	18181,82	809
Chicken Netting	5500	25,00	25000,00	597
Jackal Netting	2500	11,36	11363,64	583
Pig Netting	8000	36,36	36363,64	2474
Welded Mesh	6000	27,27	27272,73	390
TOTAL	26000	118,18	118181,82	4853

Table 4: Finished Products Calculations

6.3 Manufacturing Processes

6.3.1) Process Flowchart per Product

Figures 2–6 illustrate the process flow charts for each product individually, allowing for the identification of process overlaps.

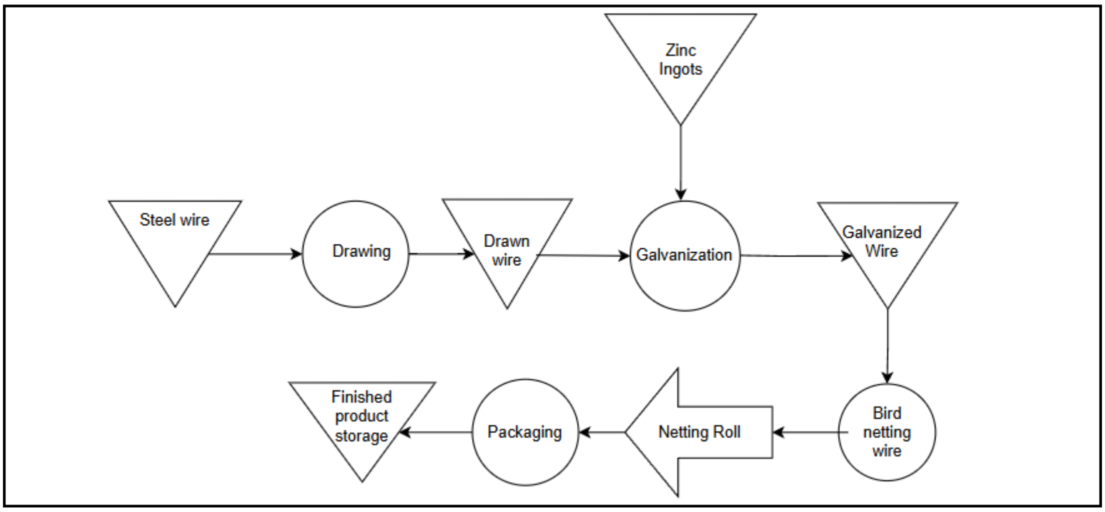

Figure 2:Bird Netting Process Map

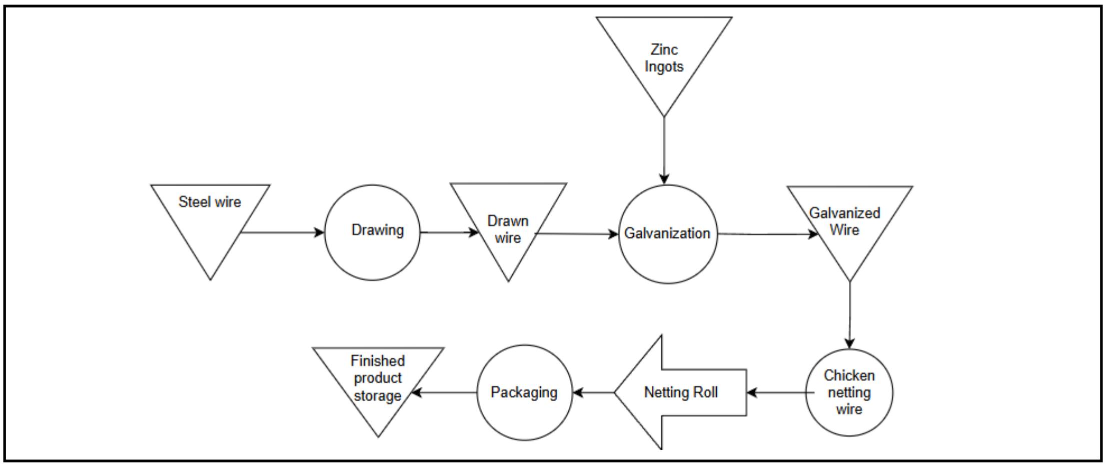

Figure 3: Chicken Netting Process Map

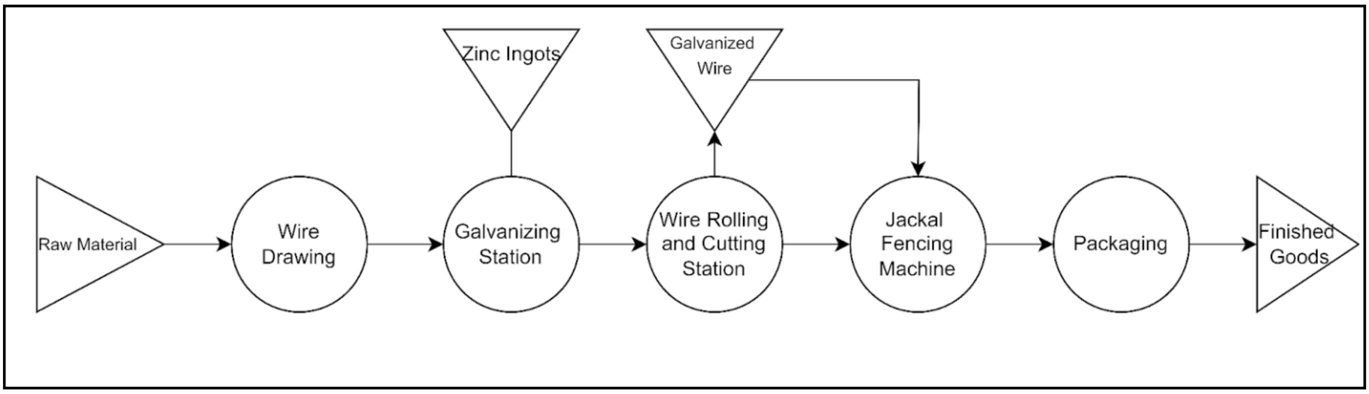

Figure 4: Jackal Netting Process Map

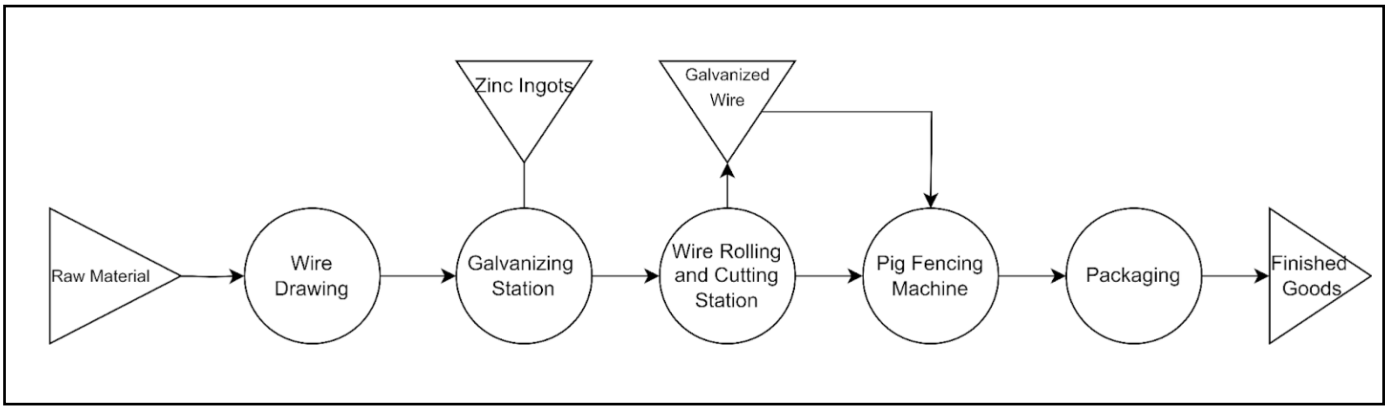

Figure 5: Pig Netting Process Map

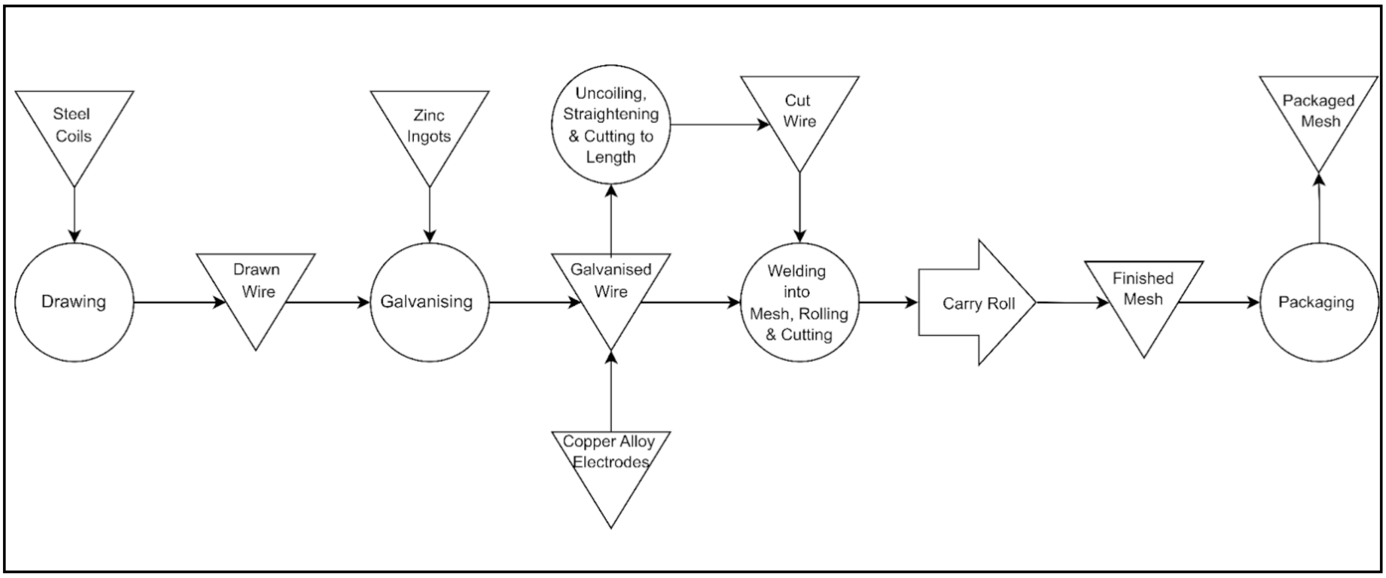

Figure 6: Welded Mesh Process Map

6.3.2) Entire Plant Process Flowchart

After identifying overlapping processes and WIP areas in the individual process flowcharts, we developed the complete production line flowchart. Figure 7 provides an overview of the entire production process, from raw steel coils to finished product storage.

Initially, raw material is galvanized and stored in the galvanized wire storage area. For Bird, Chicken, Pig, and Jackal netting, the galvanized wire is transferred to the hexagonal wire machine, where it is shaped into the correct aperture, rolled, and moved to the finished roll WIP area.

For Welded Mesh, the wires straightened, cut to length, and welded into mesh, rolled, and transferred to the finished roll WIP area. Finally, all completed rolls proceed to the packaging area before being moved to finished goods storage.

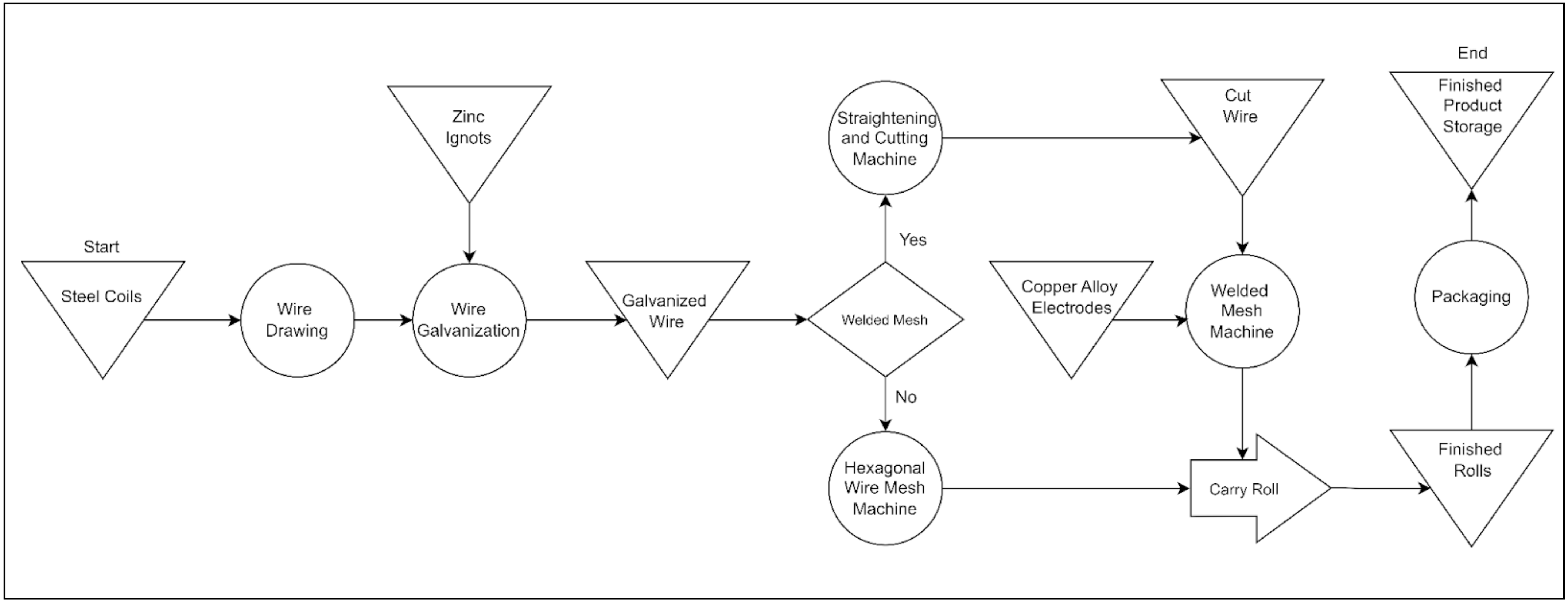

Figure 7: Manufacturing Process

6.4 Machinery, Capacity Planning and Labour

Tables 5-9 provide details of the machines selected for each process, including their specifications. The capacity planning section outlines the necessary number of machines and the required labour to operate them effectively.

6.4.1) Wire Drawing

The LZ Straight-Line Metal Drawing Machine is used to draw raw steel coils to smaller diameters, in accordance with the product specifications outlined in *Table 2*. To meet production demands, a total of 8 machines are required, while two machines are operated by one operator.

Wire Drawing	
Machine Specs	
Machine	LZ Straight-Line Metal Drawing Machine
Max. Inlet Diameter	8 mm
Outlet Diameter	0,5-4,5mm
Capacity	2 tons/hour
Speed	12m/second
Length	8600mm
Width	1390mm
Height	1925mm
Capacity Planning	
Daily Required Input (tons)	122,91
Scrap (tons)	4,73
Daily Required Output (tons) [100%]	118,18
Bird Netting[15%]	18,18
Chicken Netting [21%]	25,00
Jackal Netting [10%]	11,36
Pig Netting [31%]	36,36
Welded Mesh [23%]	27,27
Hours Available	8
Tons Requires per Hour	14,77
Machines Required (rounded up)	8
Possible Daily Output with 8 machines (tons) [100%]	128
Bird Netting[15%]	19,2
Chicken Netting [21%]	26,88
Jackal Netting [10%]	12,8
Pig Netting [31%]	39,68
Welded Mesh [23%]	29,44
Labour	
Number of Operators per Machine	0,5
Total Operators	4

Table 5: Wire Drawing Machine Calculations

6.4.2) Galvanization

A Galvanization Bath along with rollers are used to galvanize the wire. To meet production 1 machine, with 8 lines are required, while each machine is operated by two operators. To meet production demand, we require one bath that is 36 m long and 2 m wide with 8 rollers that directly pull the drawn wire from the wire drawing machines through the molten zinc.

Galvanization	
Machine Specs	
Equipment	Hotion Sink Roll and Galvanizing Bath
Wire Diameter	0.5 - 4 mm
Capacity	16 tons / hour
	128 tons / 8 hours
Roller Speed	12m/s
Length	36 m
Width	2 m
Height	1.3 m
Capacity Planning	
Daily Required Input (tons)	118.18
Daily Required Output (tons) [100%]	118.18
Bird Netting[15%]	18.036
Chicken Netting [21%]	25.2504
Jackal Netting [10%]	12.024
Pig Netting [31%]	37.2744
Welded Mesh [23%]	27.6552
Hours Available	8
Tons Requires per Hour	14.7725
Lines Required	8
Labour	
Number of Operators	2

Assumptions and Knowns		
Raw material Roll Length	7000	m
Raw Material Diameter	0.005	m
(Light)	0.03	kg/m^2
(Heavy)	0.36	kg/m^2
Rolls	50	Rolls

Zinc Ingot Calculations						
	Percentage of Roll	**Wire Diameter (m)**	**Elongation Factor (EF)**	**End Length (m)**	**Total Surface Area (m^2)**	**Total Zinc Required (kg)**
Explanation	Ratio of Product per Roll	Given	Diameter1^2 / Diameter2^2	EF x Length x Percentage	2 x pi x r^2 x l + 2 x pi x r^2	Total Surface Area x Zinc Needed
Bird Netting	15.00%	0.0006	69.4444	72916.6667	137.4447	4.1233
Chicken Netting	21.00%	0.0009	30.8642	45370.3704	128.2817	3.8485
Jackal Netting	10.00%	0.0018	7.7160	5401.2346	30.5433	0.9163
Pig Netting	31.00%	0.0018	7.7160	16743.8272	94.6841	2.8405
WeldedMesh	23.00%	0.0025	4.0000	6440.0000	50.5797	18.2087
					Zinc Required per Roll (kg)	29.9373
					Zinc Required per Day for 50 Rolls (kg)	1496.8644

Table 6: Galvanization and Zinc Calculations

6.4.3) Hexagonal Machine

The hexagonal machine is used to weave bird, chicken, pig, and jackal netting. Please note that each machine can only be set to produce one type of netting. To meet production demand, we require a total of 51 machines: 22 for bird netting, 10 for chicken netting, 6 for jackal netting, and 13 for pig netting. Each operator can operate 2 machines at once.

Hexagonal Wire Mesh Machine					
Machine Model: DP-CSR					
Length			4700 mm		
Width			1700 mm		
Height			1500 mm		
Machine Specs					
	Wire Diameter	**Mesh opening (Appeture)**	**Mesh width**	**Weaving Speed**	**Mesh per day one machine(m)**
Specs	0.4-2.2mm	Customizable (Note: One set machine can only make one mesh opening size.)	Customizable (Note: can weave 2-4 rolls of mesh at the same time.)	Assume the minimum meters per hour	Assume 8 hour shift and 4 rolls per machine at the same time
Bird Netting	0,6	13 mm	0,9 m	60-65 m/hour	1920
Chicken Netting	0,9	25 mm	1,8 m	95-100 m/hour	3040
Jackal Netting	1,8	75 mm	0,6 m	180 m/hour	5760
Pig Netting	1,8	90 mm	0,9 m	180 m/hour	5760
Capacity Planning					
	Roll Legth Required (m)	**Daily Number of Rolls Required**	**Daily Total (m) Required**	**Machines required (x)** (rounded up)	**Possible Ouput with (x) machines (rolls)** (rounded down)
Bird Netting	50	809	40450	**22**	844
Chicken Netting	50	597	29850	**10**	608
Jackal Netting	50	583	29150	**6**	691
Pig Netting	30	2474	74220	**13**	2496
Labour					
	Machines Required	**Operator per Machine**		**Total Operators**	
Bird Netting	22	0,5		11	
Chicken Netting	10	0,5		5	
Jackal Netting	6	0,5		3	
Pig Netting	13	0,5		6*	
			TOTAL AMOUNT OF OPERATORS	**19**	
* We will have a schedule where everyweek one operator needs to operate 3 machines as there isn't a even number of machines for pig netting, and because each operating only needs to ensure the machine is full and press start, one operator will be able to handle 3 machines.					

Table 7: Hexagonal Machine Calculations

6.4.4) Wire Mesh Welding Machine

The wire mesh welding machine is used to produce welded mesh. To meet production demands, we require 9 machines, each operator will operate 2 machines.

Wire Mesh Welding Machine					
Machine Model: Fully Automatic Welded Wire Mesh Making 0.8-.25mm 50-80min High Quality Welding Machine Wire Mesh Roll Spot Welding Machine					
Length			4000 mm		
Width			2500 mm		
Height			1850 mm		
Machine Specs					
	Wire Diameter	**Mesh opening (Appeture)**	**Mesh width**	**Welding Speed**	**Mesh per day per machine (m)**
Specs	2.5 - 5 mm	50 - 300 mm	Max 2 m	180 m/h	Assume 8 hour shifts
Welded Mesh	2.5	50	2 m	180 m/h	1440
Capacity Planning					
	Roll Legth Required (m)	**Daily Number of Rolls Required (rolls)**	**Daily Total (m) Required**	**Machines required (x) (rounded up)**	**Possible Ouput with (x) machines (rolls) (rounded down)**
Welded Mesh	30	390	11700	9	432
Labour					
	Machines Required	**Number of Operators per Machine**		**Total Operators**	
Welded Mesh	9	0,5		4*	
TOTAL AMOUNT OF OPERATORS				**4**	
* We will have a schedule where everyweek one operator needs to operate 3 machines as there isn't a even number of machines for welded mesh, and because each operating only needs to ensure the machine is full and press start, one operator will be able to handle 3 machines.					

Table 8: Wire Mesh Welding Machine Calculations

6.4.5) Robotic Packing System (Packaging)

The robotic packing arm is designed to automate the process of packing steel rolls on pallets. To meet demand, Wiring.com requires 13 robotic packing systems: 2 for bird netting, 2 for chicken netting, 2 for jackal netting, 6 for pig netting, and 1 for welded mesh. As one operator can manage 2 machines, a total of 7 operators will be required. (Note the exception that packing welded mesh only requires 1 machine, thus there will be one operator operating one machine.)

Robotic Packing System				
Machine Model: DP-CSR				
Length			550 mm	
Width			580 mm	
Height			2000 mm	
Machine Specs				
	Roll Diameter (Assumption) (cm)	**Rolls in one pallete**	**Packing Speed (Pallets per Hour)**	**Pallet per day per machine**
Specs	Machine can be made according to customer specifications	Refer to figure 12	60 rolls per hour	8 hour shifts
Bird Netting	30	12	5	40
Chicken Netting	30	12	5	40
Jackal Netting	30	12	5	40
Pig Netting	25	16	3,75	30
Welded Mesh	25	16	3,75	30
Capacity Planning				
	Daily Number of Rolls Required	**Total Pallets Rquired** **(rounded up)**	**Machines required (rounded up)** (Total Pallets Required / Pallets per day per machine)	**Possible Ouput with machines (pallets)**
Bird Netting	809	68	**2**	80
Chicken Netting	597	50	**2**	80
Jackal Netting	583	49	**2**	80
Pig Netting	2474	155	**6**	180
Welded Mesh	390	25	**1**	30
Labour				
	Machines Required	**Number of Operators per Machine**		**Total Operators**
Bird Netting	2	0,5		1
Chicken Netting	2	0,5		1
Jackal Netting	2	0,5		1
Pig Netting	6	0,5		3
Welded Mesh	1	1		1
			TOTAL AMOUNT OF OPERATORS	**7**

Table 9: Robotic Bagging System Calculations

6.5 Department Classification

6.5.1) Process Departments

The departments listed below are those directly involved in processing the wire. The work-in-progress (WIP) and storage areas will be considered Holding & Handling Departments, which will be discussed in the following section.

6.5.1.1) Preparation Department (1)

The purpose of this department is to process raw steel coils into wire with the correct diameter and to galvanize the wire for further production. This department will be divided into two sub-departments: one dedicated to the wire drawing machines and the other to the galvanization equipment.

- **Sub-department 1: Wire Drawing**
  This sub-department will be dedicated to the wire drawing machines as it is the first step in preparing the wire. To meet demand, we will require **4 wire drawing stations.**
  - **Wire Drawing Station:**
    Each station will consist of two wire drawing machines operated by one operator, as determined in *Table 5*.

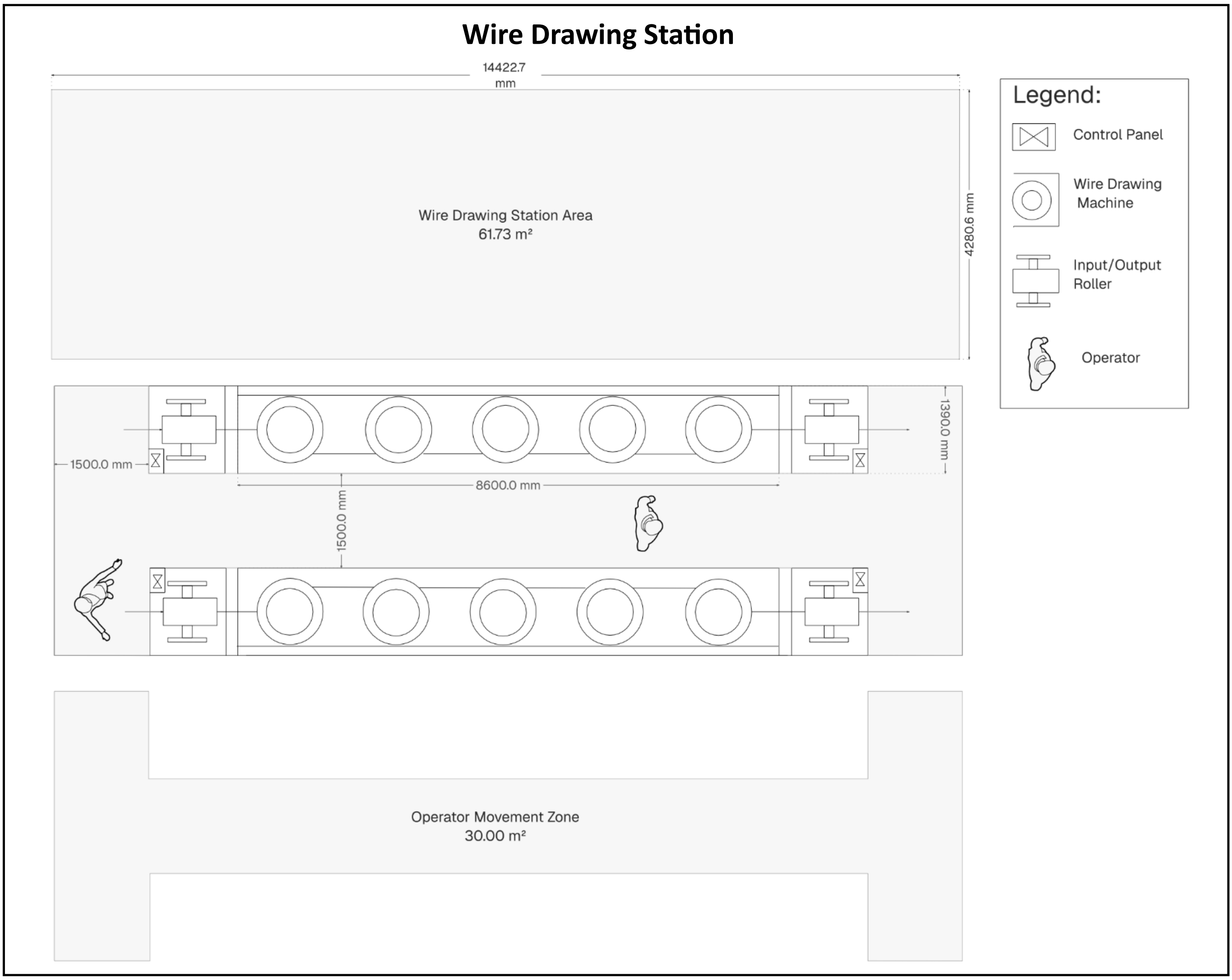

Figure 8: Station in Sub-department 1: Wire Drawing Station

- **Sub-Department 2: Galvanization**
  This sub-department will be dedicated to the galvanization process as it is the second step in preparing the wire.
    - **Galvanization Station:**
      There will be one station consisting of a big bath that is 36 m long and 2 m wide (in *figure 9* only the end of the bath is shown to fit into the page), the 8 wire drawing machines feed directly into the galvanization bath through 8 draw rollers, thus there is no WIP between the two sub-departments. The wire is directly rolled up onto spin rollers that fit onto the input roller stand (*refer to figure 10 and 11*).

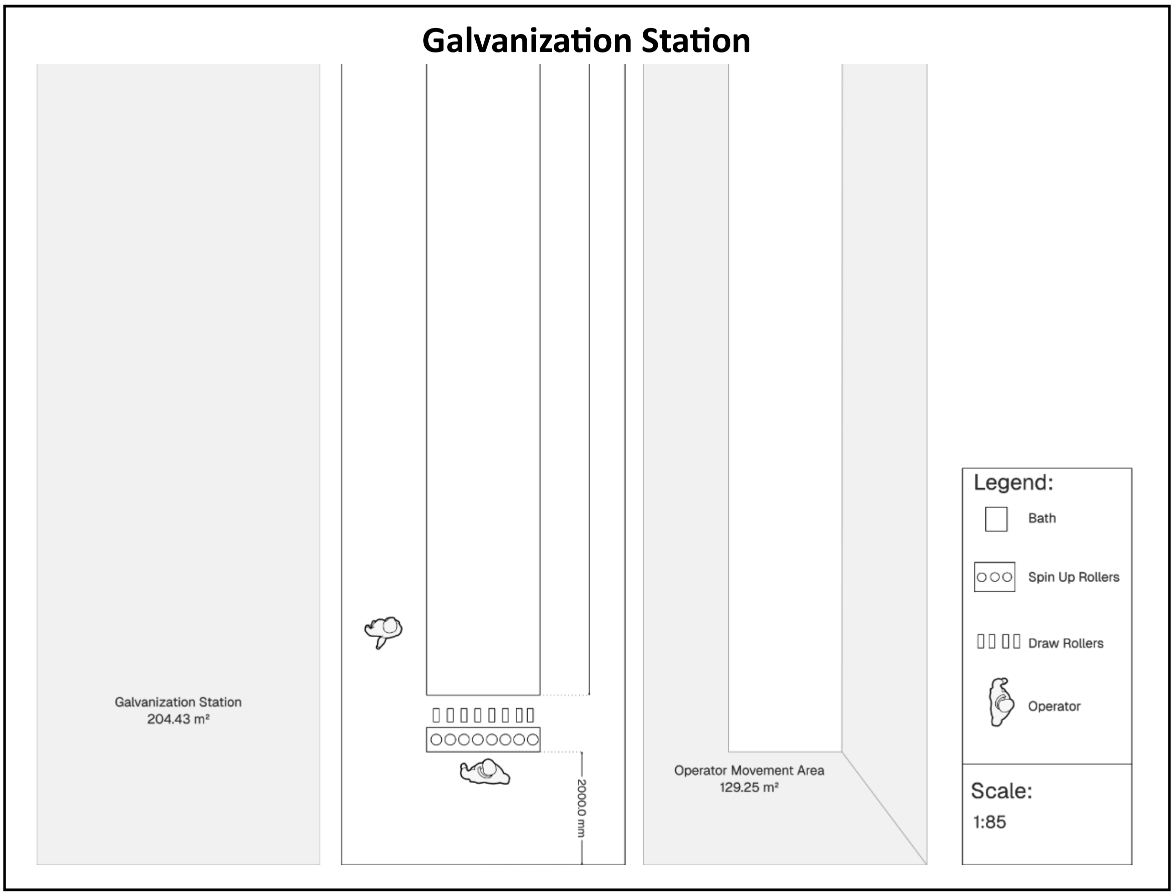

Figure 9: Galvanization Station

6.5.1.2) Hex Department (2)

The purpose of this department is to transform the drawn and galvanized wire into weaved meshed wire rolls. This department will consist of all the hexagonal wire mesh machines and will be responsible for producing rolls of bird, chicken, jackal and pig netting.

- **Sub-department 1: Hex Bird**
  This sub-department will be dedicated to all the hexagonal machines required to produce bird netting. According to table *6* we require 22 machines to meet demand. Thus, we require **11 hex stations**. Refer to the *figure 9* below.
  Sub-department 2: Hex Chicken
  This sub-department will be dedicated to all the hexagonal machines required to produce chicken netting. According to table *6* we require 10 machines to meet demand. Thus, we require **5 hex stations**. Refer to the *figure 9* below.

- **Sub-department 3: Hex Jackal**
  This sub-department will be dedicated to all the hexagonal machines required to produce jackal netting. According to table *6* we require 6 machines to meet demand. Thus, we require **3 hex stations**. Refer to the *figure 9* below.
- **Sub-department 4: Hex Pig**
  This sub-department will be dedicated to all the hexagonal machines required to produce bird netting. According to table *6* we require 13 machines to meet demand. Thus, we require **6.5 hex stations**. Refer to the *figure 9* below.

  - **Hex Station:**
    Each station will consist of 2 machines operated by one operator.

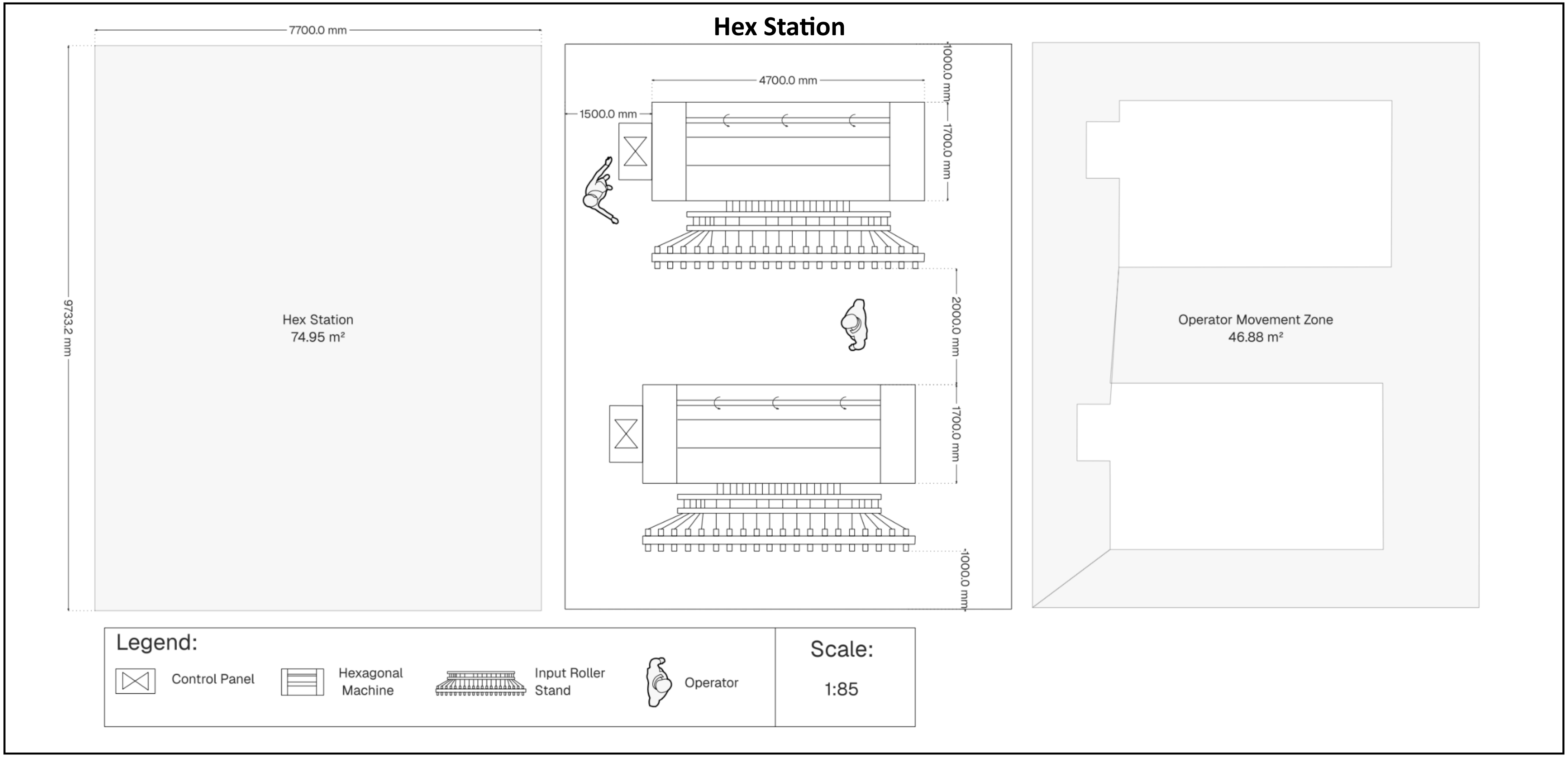

Figure 10: Hex Station

6.5.1.3) Welding Department (3)

The purpose of this department is to weld the drawn and galvanized wire into mesh rolls. This department will consist of all the wire mesh welding machines. To meet demand we require 9 machines, thus we will need **4.5 station welding stations.**

- **Welding Station:**
  Each station will consist of 2 machines operated by one operator.

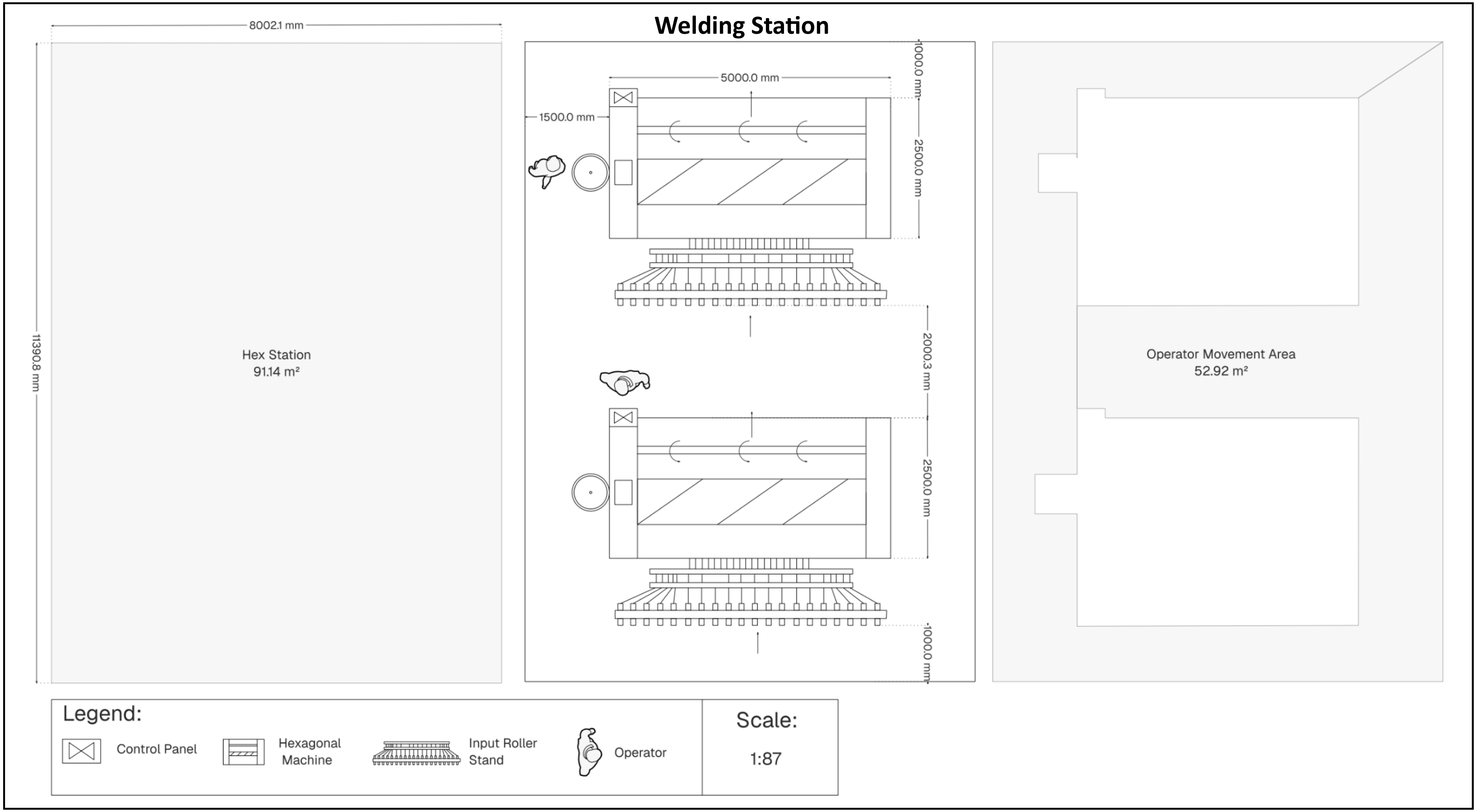

Figure 11: Welding Station

6.5.1.4) Packaging Department (4)

The purpose of this department is to pack all the rolls on pallets. This department will consist of all the robotic packaging systems that pick up the rolls and place in on the 2x2 m pallet.

The pallet is illustrated in the figure below. For bird, chicken and jackal netting the roll's diameter is 30 cm, due to the roll length being 50 m and for pig and welded mesh the roll diameter is 25 cm, due to the roll length being 30m.

Pallet Information				
	Roll Length (m)	Roll Diameter (cm) (assumed)	Pallet Visual	Rolls packed on 1 pallet
Bird	50	30	Pallet 1	12
Chicken	50	30	Pallet 1	12
Jackal	50	30	Pallet 1	12
Pig	30	25	Pallet 2	16
Welded Mesh	30	25	Pallet 2	16

Table 10: Pallet Information

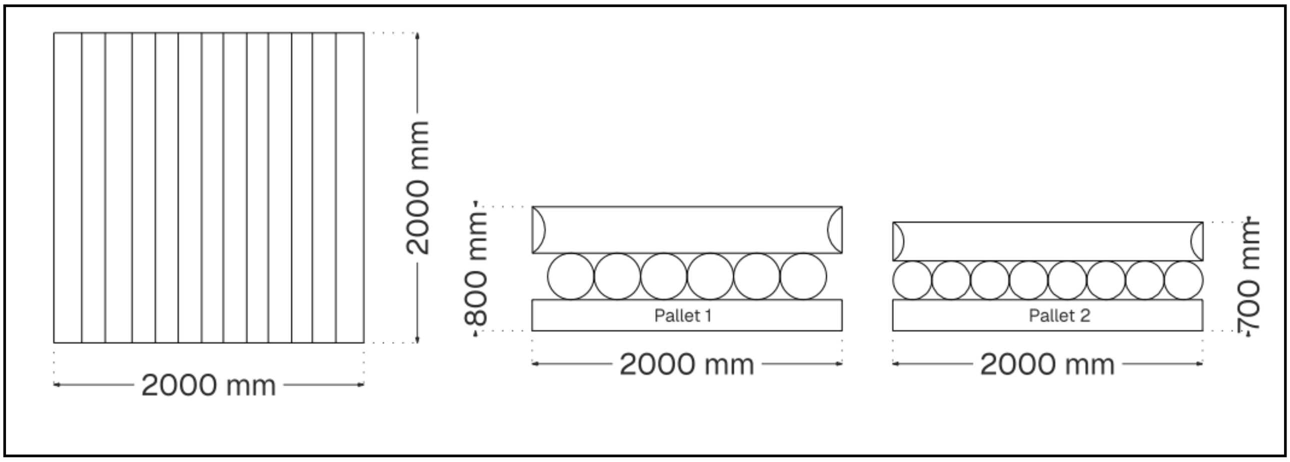

Figure 12: Pallet Visual

- **Sub-department: Bird Pack**
  This sub-department will be dedicated to all the packing machines for the bird netting. According to table 9, this netting only needs 2 machines, thus there is only **1 packing station** required.
- **Sub-department: Chicken Pack**
  This sub-department will be dedicated to all the packing machines for the chicken netting. According to table 9, this netting only needs 2 machines, thus there is only **1 packing station** required.
- **Sub-department: Jackal Pack**
  This sub-department will be dedicated to all the packing machines for the jackal netting. According to table 9, this netting only needs 2 machines, thus there is only **1 packing station** required.
- **Sub-department: Pig Pack**
  This sub-department will be dedicated to all the packing machines for the chicken netting. According to table 9, this netting needs 6 machines, thus there is **3 packing stations** required.

- **Sub-department: Welded Pack**
  This sub-department will be dedicated to all the packing machines for the welded mesh netting. According to table 9, this netting only needs 1 machine, thus is only **0.5 packing station** required.

  - **Packing Station:**
    Each station will consist of 2 machines operated by one operator.

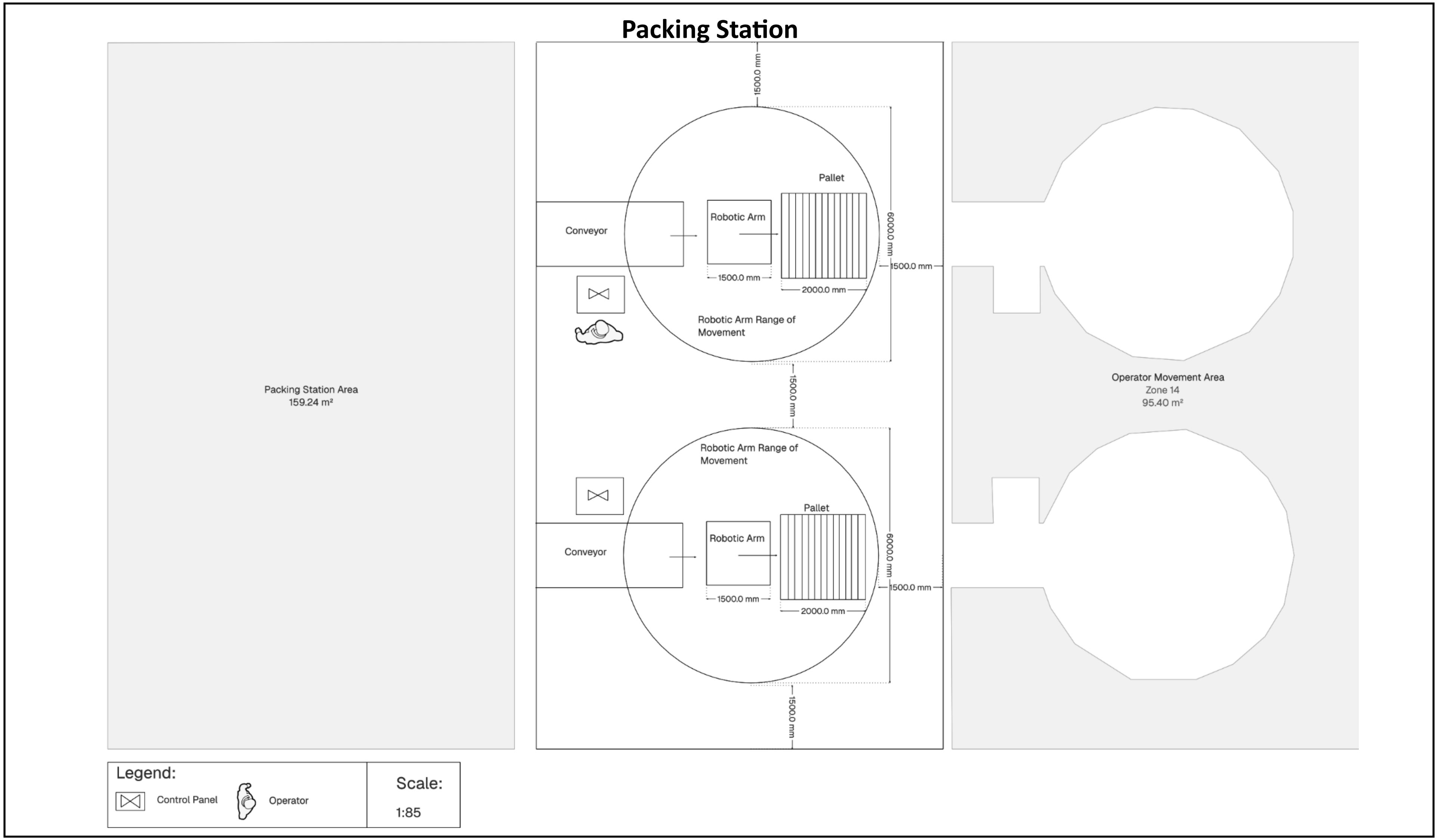

Figure 13: Packing Station

6.5.2 Visual Summary of Process Departments

Below is a visual representation of the all the process departments, with their sub-departments along with how many stations are required.

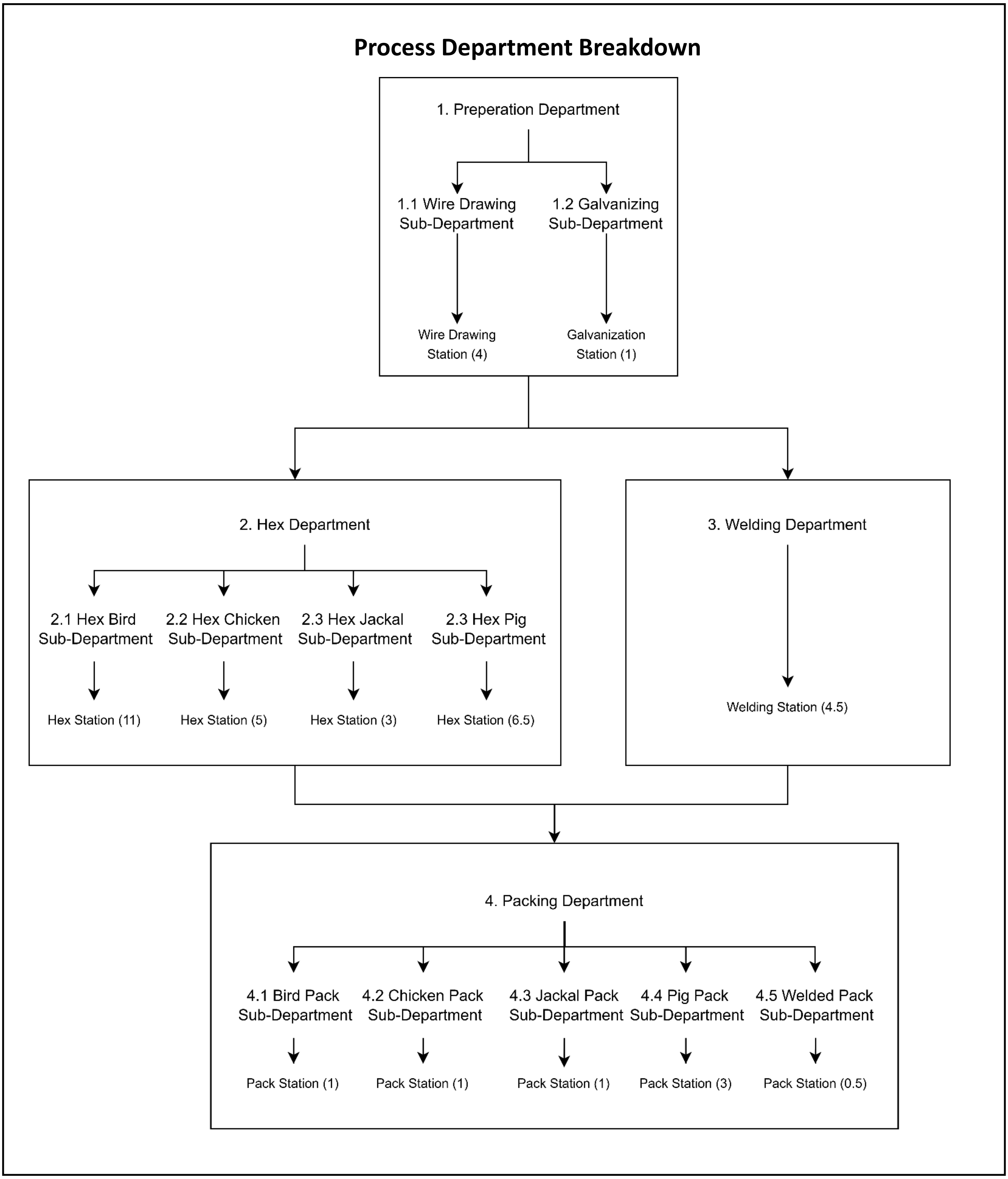

Figure 14: Process Department Breakdown

6.5.3) Holding & Handling Departments

The departments listed below are not directly involved in wire processing but are responsible for its transportation and storage between different areas.

6.5.3.1) Raw Material Storage

The purpose of this station is to store raw materials, which are steel coils.
First In First Out system is used to allow easy access to raw materials and movement between departments.
Coils will be stacked on top of each other up to 3 levels.
Movement of delivered coils into the warehouse is done by a forklift.
There is an aisle in between rows to allow movement of the forklift.

Figure 15: Forklift

Specifications:

Name	2025 REVARO T-REX 830
Load capacity	3000 kg
Turning radius	3.5 m
Drive	4WD

Table 11: Forklift Specs

Storage: KLP Roll stop system
The KLP RollStop System is a renowned solution for the safe and flexible storage of steel coils, widely adopted by major steel companies globally. This turn-key system offers both maximum flexibility and safety, allowing coils to be stored up to three levels high.

Key Components:
The system comprises four primary components:

1. Rail: A reinforced base with notches to position the RollStops.
2. Connector: Joins rails to achieve the desired row length.
3. Spacer: Sets the appropriate row width.

4. RollStop: Adjustable stops that accommodate various coil diameters.

This modular design ensures straightforward on-site installation.

Advantages:

- Flexibility: Accommodates a wide range of coil diameters and widths.
- Safety: Enhances personnel safety and reduces coil damage; TÜV-approved.
- Durability: Constructed from robust polymeric materials, the system is virtually indestructible and splinter-proof.
- Ease of Installation: Requires no special equipment; two individuals can complete the setup efficiently.
- Stacking Capability: Supports easy stacking of coils up to three levels.
- Low Maintenance: Resistant to oil absorption and designed for a long lifespan.

Figure 16: KLP rollstop system

Number of roll coils ordered is 150 and the coils are stored in the raw materials warehouse for 3 days and used before new stock arrives.

Material Handling	
Assumptions & given data	
Roll Height (m)	1
Roll diameter (m)	1.5
Radius	0.75
Radius^2	0.5625

Roll weight (kg)	2500
Pi	3.141592654
number of rolls	150
Raw materials warehouse	
Area of 1 coil	8.246680717
Foot Area m^2	412.3340358
Forklifts	2

Table 12: Assumptions and Given data for material handling of Raw Material

Moving Raw materials to and from the warehouse is done by an overhead crane designed by RGM.
Specifications:
engineer a SANS 10375:2018-compliant crane system to meet your requirements:

Name	RGM single girder overhead crane
Span	35 meter (can be increased)
Load capacity	10 tons
Long Travel speed	80m/min
Cross travel speed	40 m/min
Hoist speeds	5 to 20 m/min

Table 13: Specs of Crane

6.5.3.2) Scrap Storage

The wire drawing process has a scrap rate of 4%. Below is the storage capacity of the wire scrap over three days.

Total scrap (kg)	4730
3days	14190
Area calculated based on Assumption	
Component	**Specification**
Storage Method	Outdoor paved yard with containment

Height Limit	2m (safe stacking height)
Capacity	71 m²
Security	Fenced + covered area
Access	Forklift accessible

Table 14: Storage Capacity of Scrap

6.5.3.3) Prepared Wire Storage (WIP)

Hex Department											
	Daily Number of Rolls Required (rolls)	**Roll Diameter (m) Assumptions**	**Roll Height (m)**	**Rack size (m x m)**	**Number of rolls a rack can hold**	**Number of Stackings**	**Maximum height (m)**	**Number of rolls**	**Racks needed**	**Area (m^2)**	**Number of workers**
Bird Netting	809	0.30	0.90	1.00	9.00	3	3	27	30	30	4
Chicken Netting	597	0.30	1.80	1.80	36.00	2	3.6	72	9	30	5
Jackal Netting	583	0.30	0.60	1.80	108.00	1	1.8	108	6	20	4
Pig Netting	2474	0.25	0.90	1.80	51.84	2	3.6	103.68	24	78	6
Welding Department											
Welded Mesh	390	0.25	2.00	2.00	64.00	1	2	64	7	28	3
										total	22

Table 15: Prepared Wire Storage

6.5.3.4) Processed Wire Storage (WIP)

Conveyor Loading Calculations						
	Daily Number of Rolls Required (rolls)	**Machines required (x) (rounded up)**	**Daily Output per Machine (Rolls)**	**Hourly Output per Machine (Rolls/h)**	**One Roll Production Time (min)**	**Workers Required (1 worker can place a roll every 3 min)**
Bird Netting	809	22	36.77	4.60	13.05	5
Chicken Netting	597	10	59.70	7.46	8.04	3
Jackal Netting	583	6	97.17	12.15	4.94	2
Pig Netting	2474	13	190.31	23.79	2.52	1
Welded Mesh	390	9	43.33	5.42	11.08	4
Total Number of Workers Required						15

Table 16: Conveyor Loading Calculations

Hex Department to Packaging Department:

Once the hexagonal netting machine finishes producing a roll, it is loaded onto a low-lying roller conveyor that leads to the packaging department. This allows the workers to transport the rolls from the hex department to the packaging department with minimal physical effort.

Additionally, the conveyor acts as WIP storage, ensuring a buffer area for when mesh production surpasses packaging speed. When a robot arm at the packaging station is done packing a roll, it can simply pick up the next one from the WIP conveyor.

All 51 netting machines will need a conveyor leading from it. The 51 conveyors will all converge into thirteen groups of three or four conveyors. At these intersections, a worker will be needed to guide the rolls away from each other.

Operators between Hex and Packaging Departments	
	Number of Operators
Loading the Conveyors	11
Intersection Operators	7
Oversight	1
Total	19

Table 17:Operators between Hex and Packing Station

Welded Mesh Department to Packaging Department:

Once the welded mesh machine finishes producing a roll, it is loaded onto a low-lying roller conveyor that leads to the packaging department. This allows the workers to transport the rolls from the hex department to the packaging department with minimal physical effort.

Additionally, the conveyor acts as WIP storage, ensuring a buffer area for when mesh production surpasses packaging speed. When a robot arm at the packaging station is done packing a roll, it can simply pick it up from the WIP conveyor.

All nine welded mesh machines will need a conveyor leading from it. The 9 conveyors will all converge into three groups of three conveyors. At these intersections, a worker will be needed to guide the rolls away from each other.

Operators between Mesh and Packaging Departments	
	Number of Operators
Loading the Conveyors	5
Intersection Operators	2
Oversight	1
Total	8

Table 18:Operators between Mesh and Packaging

6.5.3.5) Finished Goods Storage

Packaging Department to Finished Goods Storage:

The Robot arms receive the finished product via the conveyors from the hex and welding departments. As the finished wire rolls are picked up, the robotic claw wraps the packaging around it. It is then lifted and placed onto a pallet in a pre-determined position to maximise stability and the number of rolls you can fit onto it. Once the pallet is packed, it is wrapped with a poly woven pallet strap to keep the fence rolls from falling off.

After the pallets are secured, an overhead crane is used to move the pallets from the packing department to the finished goods storage.

Operators Between Packaging and Storage Departments	
	Number of Operators
Crane Coordinator	1
Stock Manager	1
Total	2

Table 19: Operators between packaging and storage department

Finished Goods Area:

The total area for the finished goods is calculated by multiplying the area needed for a pallet with the number of pallets needed to meet the daily demand.

Finished Goods Area			
	Number of Pallets (2 days)	**Pallet Size (m^2)**	**Area Needed (m^2)**
Bird Netting	**136**	**4**	**544**
Chicken Netting	**100**	**4**	**400**
Jackal Netting	**98**	**4**	**392**
Pig Netting	**310**	**4**	**1240**
Welded Mesh	**50**	**4**	**200**
Total Area Needed			**2776**

Table 20: Finished Goods Storage Area

6.5.4) Supporting Activities and Areas

6.5.4.1) Power Usage Calculations:

Power consumption was calculated using average power ratings for all the listed equipment. The power ratings were then multiplied by the quantity of equipment and the operating hours.

Power Calculations						
Department	**Equipment/Process**	**Power Rating (kW)**	**Quantity**	**Total Power (kW)**	**Operating Hours per Day**	**Daily Energy Consumption (kWh)**
Raw Material Storage	**Lighting**	**0.15**	**12**	**1.8**	**9**	**16.2**
	Ventilation	**1.15**	**2**	**2.3**	**9**	**20.7**
	Overhead Crane	**27500**	**1**	**27500**	**8**	**220000**
Wire Drawing & Galvanizing	**Lighting**	**0.15**	**12**	**1.8**	**9**	**16.2**
	Ventilation	**1.15**	**5**	**5.75**	**9**	**51.75**
	Wire Drawing Machines	**37**	**8**	**296**	**8**	**2368**
	Galvanizing Bath Heaters	**25000**	**1**	**25000**	**8**	**200000**

	Exhaust & Fume Extraction	**1.5**	**2**	**3**	**8**	**24**
Hex Machine Department	**Lighting**	**0.15**	**12**	**1.8**	**9**	**16.2**
	Ventilation	**1.15**	**2**	**2.3**	**9**	**20.7**
	Hexagonal Weaving Machines	**4**	**51**	**204**	**8**	**1632**
	Conveyors	**0.745**	**51**	**37.995**	**8**	**303.96**
Welded Mesh Department	**Lighting**	**0.15**	**12**	**1.8**	**9**	**16.2**
	Ventilation	**1.15**	**2**	**2.3**	**9**	**20.7**
	Welding Machines	**1920**	**9**	**17280**	**8**	**138240**
	Conveyors	**0.745**	**9**	**6.705**	**8**	**53.64**
Packaging	**Lighting**	**0.15**	**12**	**1.8**	**9**	**16.2**
	Ventilation	**1.15**	**2**	**2.3**	**9**	**20.7**
	Strapping Machines	**0.5**	**14**	**7**	**8**	**56**
	Robot Arms	**1.5**	**14**	**21**	**8**	**168**
Finished Goods Storage	**Lighting**	**0.15**	**12**	**1.8**	**9**	**16.2**
	Ventilation	**1.15**	**2**	**2.3**	**9**	**20.7**
	Overhead Crane	**27500**	**1**	**27500**	**8**	**220000**
Offices	**Lighting**	**0.03**	**5**	**0.15**	**10**	**1.5**
	Ventilation	**0.024**	**5**	**0.12**	**10**	**1.2**
Outside Utilities	**Floodlights**	**0.05**	**10**	**0.5**	**14**	**7**
Total Power Consumption						**783107.75**

Table 21: Power Calculations

6.5.4.2) Lights

Preliminary estimates suggest that 12 LED lights per department, 1 LED light per office, and 10 floodlights outside the building should be sufficient.

Lighting Calculations		
	Amount Needed	**Type**
Raw Material Storage	**12**	**Industrial LED**
Wire Drawing & Galvanizing	**12**	**Industrial LED**
Hex Machine Department	**12**	**Industrial LED**
Welded Mesh Department	**12**	**Industrial LED**
Packaging	**12**	**Industrial LED**
Finished Goods Storage	**12**	**Industrial LED**
Offices	**5**	**Office LED**
Outside	**10**	**LED Floodlight**

Table 22: Lighting Calculations

6.5.4.3) Amenities

1. **Personnel**

17 Management personnel oversee the day-to-day activities of the manufacturing plant

Management Personnel	
Personnel	Number of people
Plant Management & Operations	5
Plant manager	1
Operations supervisors	4
Sales & Customer Relations	3
sales & marketing manager	1
sales representatives	2
Purchasing & Inventory Management	1
Manager	1
Finance & Administration	5
Finance manager	1

Accountants	2
Administrative Staff	2
Maintenance	1
Supervisor	1
Logistics & Dispatch	2
Logistics manager	1
Warehouse/ dispatch Supervisor	1

Table 23: Management Personnel

Number of operators and supporting staff that deals with material handling

Daily operations personnel	Operators	Material handling
Wire drawing	8	4
Galvanization	2	2
Hex department	51	19
Weld department	9	3
Packaging and packing	7	35

Table 24: Daily Operators

2. Locker rooms

Total area of the locker rooms which include rest rooms. 83 double-tier locker rooms that can accommodate 165 employees which is the operators and the supporting staff.

Locker rooms with Rest rooms			
Assumptions and Knowns			
Employees	165		
Area of Locker rooms(m^2)	0.135		
Total floor space	22.275		
Double tier locker	11.1375		
Component	**Area (m^2)**	**Quantity**	**notes & details**

Double locker	11.1375	83	units
Ailes	15	1.2	meter between rows
Changing area	20		open space for changing
Benches			Include in locker rows
Toilet	15.6	13	5 male 7 female
Urinals	4	5	male
Sinks	7.7	11	
Total	73.4375		

Table 25: Locker Rooms

3. Cafeteria

A 50-seat cafeteria with an area of 125 m^2 for providing basic food.

Cafeteria		
Area	**Size (m²)**	**Details**
Dining Area	125	50 seats @ 2.5m²/seat
Food Service Counter	15	8m long serving line
Kitchenette	25	Basic food prep/storage
Vending Area	8	4 vending machines
Storage/Circulation	20	Supplies and movement space
Total	**193**	

Table 26:Cafeteria

5. **Reception area**

Reception offices, to receive customers and suppliers bringing the paperwork for deliveries.

Reception area & Restroom for offices		
Component	**Size (m²)**	**Details**
Waiting Area	20	8 seats + coffee table
Reception Desk	10	L-shaped counter with storage

Visitor Lockers	5	Secure storage for personal items
Admin Workstation	15	For reception staff
Circulation	10	Clear pathways
Total	**60**	

Table 27: Reception Area

Appendix

Will add everything – how tables were calculated etc.